



Fake missile warning by Nest cam

Hackers sent fake warnings of incoming North Korea missile attack on California through family's Nest Cam. <https://digit.in/feb19-15>



DxOMark selfie is here

After a decade in camera testing, DxOMark will test front-facing camera performance in smartphones. <https://digit.in/feb19-16>

Sapphire NITRO+ Radeon RX 590 Special Edition Graphics Card

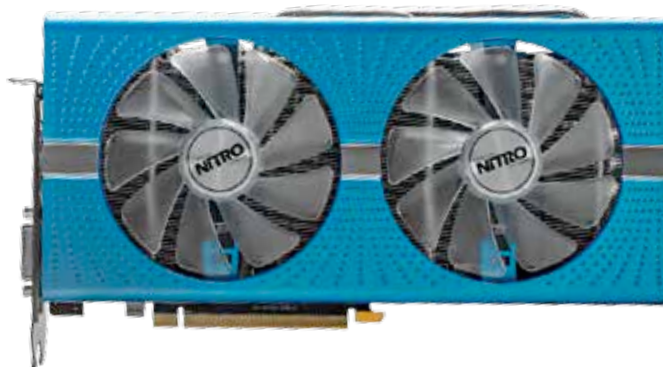
We've seen this one before

Roughly two-and-a-half years later, we're looking at the same Polaris cards even today. Polaris itself was a marked improvement over the previous generation of Radeon graphics cards and the roadmaps showcased an upward trend but we're yet to see a worthy successor to the Polaris lineup.

The Radeon RX 590 is another revision of the RX 480. The RX 480 was followed by the RX 580 which had the same GPU configuration but sported slight improvements that allowed it to scale to slightly higher frequencies. However, the board power requirement also went up to 185 watts from 150 watts that the RX 480 was designed for. In a similar vein, we now have the RX 590 which is a slightly improved version of the RX 580.

What's new?

The clock speeds have improved a lot and the typical board power values have also risen to match the clock speeds. The other key element that stands out is the move to a smaller process node. This has been at the centre of much debate with folks claiming that the GLOBALFOUNDRIES 12nm process node is simply an interpretation of Samsung's 14nm process node. So it is actually 12nm? Well, none of the transistor sizes are actually what they seem to be. The actual values of the different components of the transistor are much larger than 12 or 14nm. Fabs simply call it 12nm or 14nm to keep things simple. The differences between two manufacturing processes are vast. The fact here is that the GLOBALFOUNDRIES 12nm is an improvement over the 14nm process node and that's why the RX 590 can clock higher. It definitely will consume more power which is why the TBP for



Price
₹26,900

the RX 590 is now 225 Watts. Simply bumping up the power to the board wouldn't have allowed the average user to achieve these clocks.

Performance

In all our synthetic benchmarks, the Sapphire NITRO+ Radeon RX 590 is evenly matched with the NVIDIA GTX 1060 6GB. Since few benchmarks can actually flood the frame buffer, we don't see any difference in scores. Most 3DMark benchmarks at their default configurations also run at FHD or lower resolutions. The newer gen RTX 2060 that came out recently is ahead of both cards. However, it also is a lot more expensive than the RX 590 and the GTX 1060, so it's not really an apples to apples comparison.

In all our video game benchmarks, the Sapphire NITRO+ Radeon RX 590 has a clear lead over the older NVIDIA GTX 1060 6GB graphics card. Except for The Witcher 3, in all other games the Sapphire NITRO+ Radeon RX 590 gets higher frame rates at 2160p and 1440p. Compared to the recently released RTX 2060, both the mid-range cards appear at a disadvantage.

Pricing

The Sapphire NITRO+ Radeon RX



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590 is priced at ₹26,900 on some of the popular computer specialist stores. What sweetens the deal is that AMD is currently offering three AAA games with the Sapphire NITRO+ Radeon RX 590, not one but three. Each worth ₹2,999. If we were to offset this cost

from the graphics card then we're paying ₹17,900 for this particular SKU and all other partner cards which are participating in the bundle offer.

Verdict

The RX 590 is an interesting graphics card. It's better than the stock GTX 1060 but might probably be bested by an overclocked GTX 1060. However, getting the same chip out for three years in a row seems a stretch, or folks wondering if they should get an RX 590 for mid-range gaming, you most certainly should given the fact that the included bundle makes this one of the sweetest deals you'll get for a while.

—Mithun Mohandas

SPECIFICATIONS

STREAM PROCESSORS: 2304 | CU: 36 | BOOST CLOCK: 1560 MHz
| VRAM: 8 GB GDDR5 | BUS-WIDTH: 256-bit | CONNECTORS:
1x8PIN+1x6-PIN | TBP: <250W

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